TRAINER-READY END-TO-END CASE STUDY

Prescriptive Analytics Case Study

Automobile Predictive Maintenance and Service Optimization

Architecture, Python Solution, Constraints, Reports, Tests and Governance

Prepared: June 2026

Contents

Use this section as the recommended reading order for the document.

1. Prescriptive Analytics - 5W1H

Question	Explanation
What?	Prescriptive analytics recommends the best action after considering predictions, rules, costs, constraints, and business objectives.
Why?	A prediction such as “89% failure risk” does not itself prevent a breakdown. The business needs to know what to do, when to do it, what resources are required, and what to do when capacity or parts are unavailable.
Who?	Service operations managers, workshop planners, data scientists, customer-care teams, inventory planners, fleet managers, and compliance or safety teams.
When?	After descriptive analysis explains the past and predictive analytics estimates the future, especially when a decision must be made under limited time, budget, people, capacity, or inventory.
Where?	Predictive maintenance, healthcare treatment planning, supply-chain replenishment, workforce scheduling, pricing, fraud response, route optimization, and customer-retention programs.
How?	Combine risk predictions with decision rules, expected-value calculations, optimization or prioritization, operational constraints, action automation, and feedback monitoring.

Primary question answered: What should the organization do next?

2. Business Problem

A vehicle manufacturer and its service network want to reduce unexpected breakdowns, emergency repairs, customer complaints, towing costs, and safety incidents. The organization already collects vehicle and service data and can estimate the probability of failure in the next 30 days.

The missing capability is an operational decision system. For each vehicle, it must decide:

1. Whether intervention is required.
2. The safest and most economical action.
3. How urgently the action must be completed.
4. Which workshop should handle it.
5. Which spare parts must be reserved.
6. Whether workshop capacity and parts inventory make the action executable.
7. What escalation is required if resources are unavailable.
8. How the result should be communicated and measured.

3. Business Objectives and Success Criteria

Objective	Measurement

Reduce unexpected breakdowns	Breakdown rate among contacted high-risk vehicles
Prioritize safety-critical vehicles	Percentage of safety overrides actioned within 24 hours
Use workshop capacity effectively	Reserved slots divided by available urgent slots
Reduce parts-related delays	Parts escalations resolved before scheduled appointment
Improve customer experience	Contact success rate, appointment acceptance, complaint reduction
Demonstrate financial value	Expected and realized savings after intervention cost
Maintain trustworthy decisions	Audit trail, data-quality pass rate, policy compliance, model monitoring

Acceptance targets for the case study

- Every valid vehicle receives a risk category, recommended action, timeframe, priority score, and explanation.
- Safety overrides take precedence over normal risk thresholds.
- Near-term service slots are assigned in priority order.
- Spare-parts stock is consumed as jobs are reserved.
- Unavailable parts produce an explicit parts escalation.
- Exhausted workshop capacity produces a capacity escalation.
- All decisions are exported to an action report and summarized for management.

4. Stakeholders and Responsibilities

Stakeholder	Responsibility
Service operations manager	Owns service policy, capacity rules, and execution KPIs
Data science team	Maintains the failure-risk model and probability quality
Workshop planner	Confirms slots, technicians, and alternate workshops
Inventory planner	Reserves, transfers, or procures required parts
Customer-care team	Contacts customers and records appointment response
Safety/compliance team	Defines non-negotiable safety overrides and audit controls
IT/data engineering	Maintains ingestion, APIs, data quality, security, and monitoring
Management	Reviews financial value, risk reduction, and service-network performance

5. Data Inputs

5.1 Vehicle and customer data

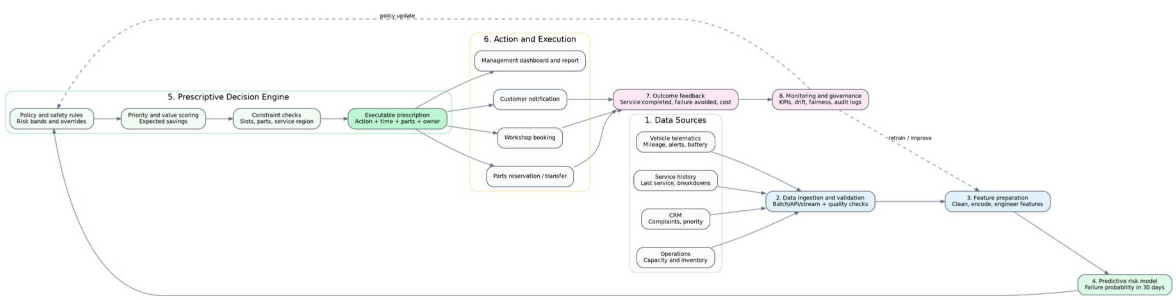
Feature	Meaning	Example
mileage_km	Total vehicle mileage	72,000
service_gap_months	Months since last service	14

engine_alerts	Recent engine warning count	5
brake_wear_percent	Estimated brake wear	82%
battery_voltage	Battery condition	Low
customer_complaints	Recent complaint count	3
vehicle_age_years	Age of vehicle	6.5
breakdown_history	Previous breakdown count	2
customer_priority	Standard, Fleet, or VIP	Fleet
workshop_region	Nearest service region	Pune West
actual_failure_30d	Historical training target	0 or 1

5.2 Operational constraints

- Workshop slots available today and during the next 48 hours.
- Mobile-technician capacity.
- Regional stock of brake kits, batteries, and engine sensors.
- Service-region ownership and alternate-workshop escalation.
- Safety rules that cannot be overridden by cost optimization.

6. Architecture Flow



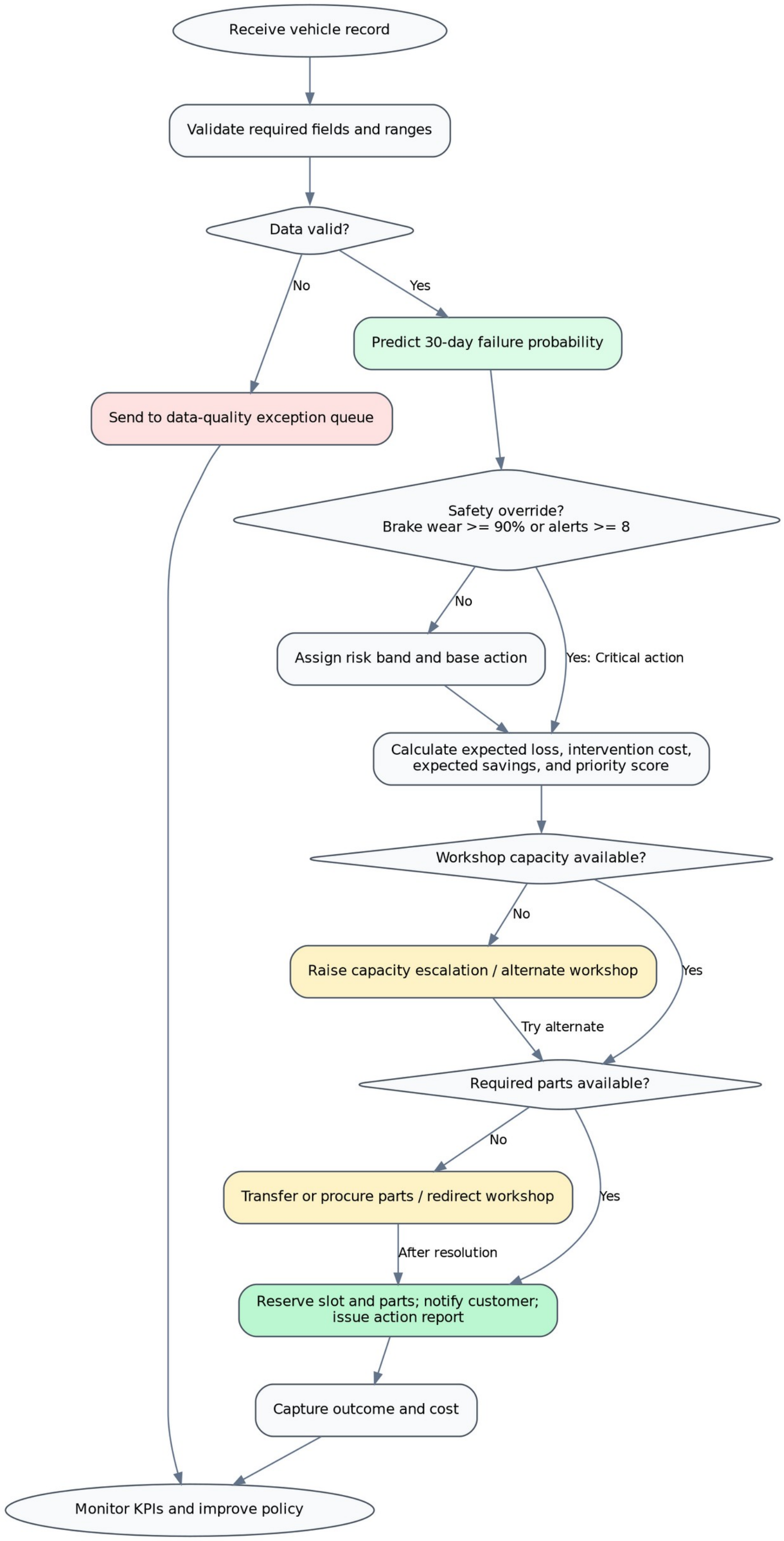
Prescriptive analytics architecture

Architecture explanation

1. **Data sources:** telematics, service history, CRM, workshop capacity, and parts inventory.
2. **Ingestion and validation:** batch, API, or streaming data is checked for missing values, invalid categories, impossible ranges, and duplicate identifiers.
3. **Feature preparation:** numerical values are scaled and categorical values are encoded.
4. **Predictive risk model:** produces a probability of failure in the next 30 days.
5. **Prescriptive engine:** applies safety policies, calculates expected value, ranks cases, and checks capacity and inventory constraints.
6. **Action systems:** customer notification, booking, parts reservation, transfer or procurement, and management reporting.
7. **Feedback:** service completion, actual failures, cost, and customer response are captured.

8. **Governance:** model performance, policy compliance, drift, fairness, data quality, and audit logs are monitored.

7. Decision Workflow



Prescriptive decision flow

Decision sequence

1. Validate the incoming record.
2. Predict failure probability.
3. Check safety overrides.
4. Assign risk band and base action.
5. Estimate loss without action and intervention cost.
6. Calculate expected savings and priority score.
7. Check workshop capacity.
8. Check parts availability.
9. Reserve the slot and parts or raise an escalation.
10. Notify the customer and publish the action report.
11. Capture the outcome and use it to improve the model and policy.

8. Prescriptive Decision Rules

Condition	Risk / priority	Prescribed action	Target time
Brake wear >= 90% or engine alerts >= 8	Critical / 1	Immediate safety intervention	Within 24 hours
Failure probability >= 0.80	Critical / 1	Immediate safety intervention	Within 24 hours
Probability 0.65-0.7999	High / 2	Priority workshop inspection	Within 48 hours
Probability 0.40-0.6499	Medium / 3	Planned preventive service	Within 7 days
Probability 0.20-0.3999	Low / 4	Remote diagnostic and monitoring	Within 30 days
Probability < 0.20	Low / 5	No immediate intervention	Next regular service

Parts-reservation rules

Observed condition	Part to reserve
Brake wear >= 75%	Brake Kit
Battery voltage = Low	Battery
Engine alerts >= 4	Engine Sensor

Important rule precedence

A safety override is evaluated before normal risk thresholds. Therefore, a vehicle may receive a **Critical** action even when its model probability is below 0.80. This avoids allowing a statistical model to overrule a clear safety condition.

9. Business-Value Calculation

The engine compares the expected cost of doing nothing with the expected cost after intervention.

```
Expected loss without action
    = Predicted failure probability x Estimated breakdown cost

Expected loss after action
    = Residual failure probability x Estimated breakdown cost
      + Intervention cost

Expected savings
    = Expected loss without action - Expected loss after action
```

The demonstration uses different effectiveness assumptions by action urgency. These are case-study assumptions and must be replaced with validated historical effectiveness values before production use.

Priority score

The priority score combines:

- Failure probability.
- Safety bonus.
- Customer-priority bonus.
- Complaint bonus.
- Brake-wear severity.

The score is used only after the non-negotiable priority rank. A Priority 1 case is always considered before Priority 2, even if the Priority 2 score is numerically high.

10. Resource Optimization Logic

The case study uses a transparent greedy allocation approach:

1. Group vehicles by workshop region.
2. Sort by priority rank and then priority score.
3. Check whether required parts are available.
4. Allocate today's slots first.
5. Allocate next-48-hour slots when today's capacity is exhausted.
6. Decrement parts stock after every confirmed reservation.
7. Raise a parts escalation when stock is unavailable.
8. Raise a capacity escalation when all urgent slots are used.

This approach is explainable and suitable for training. A production system may replace it with mathematical optimization when there are more complex constraints such as technician skills, bay type, travel time, appointment preferences, service duration, or multi-period inventory.

11. Worked Vehicle Example

The package selects a representative case from the generated data.

Item	Value
Vehicle ID	VEH-0469
Mileage	85,650 km
Service gap	11.7 months
Engine alerts	3
Brake wear	82.7%
Battery	Normal
Complaints	2
Customer priority	VIP
Workshop	Pune West
Predicted failure probability	77.37%
Risk category	High
Prescribed action	Priority workshop inspection
Target time	Within 48 hours
Required parts	Brake Kit
Scheduled window	Today
Execution status	Slot reserved
Estimated intervention cost	INR 13,500.00
Expected savings	INR 40,357.71

Interpretation

The model predicts a high probability of failure. Brake wear also requires a brake kit. The prescriptive engine assigns a Priority 2 workshop inspection, confirms parts availability, reserves a slot today, and estimates the financial benefit of acting before a breakdown occurs.

12. Demonstration Results

The supplied dataset contains **500 vehicles**.

Predictive model results

Metric	Result
Test accuracy	0.736
ROC-AUC	0.785
Test confusion matrix	[[73, 10], [23, 19]]

The model is included to supply risk probabilities for the prescriptive layer. Its metrics are demonstration results from synthetic data, not production performance claims.

Prescriptive output results

KPI	Result
Critical and High vehicles	96
Slots reserved	118
Parts escalations	73
Capacity escalations	50
Total modeled expected savings	INR 8,278,817.77

Risk summary

Risk category	Vehicles	Avg. failure probability	Expected savings (INR)
Critical	44	68.7%	1,522,772.22
High	52	71.4%	2,020,322.76
Medium	145	51.2%	3,082,299.10
Low	259	24.3%	1,653,423.69

Workshop-region summary

Workshop region	Vehicles	Critical/High	Slots reserved	Parts escalations	Capacity escalations	Expected savings (INR)
Mumbai	92	20	30	0	13	1,544,393.33
Pune Central	124	21	24	28	2	1,837,839.06
Pune North	136	29	28	31	10	2,399,801.04
Pune West	148	26	36	14	25	2,496,784.34

13. Required Reports

13.1 Vehicle action report

One row per vehicle containing:

- Predicted probability and risk category.
- Recommended action and timeframe.
- Priority rank and score.
- Required parts and availability.
- Scheduled window and execution status.
- Estimated costs and expected savings.
- Human-readable decision reason.

13.2 Management dashboard

The workbook contains:

- Executive KPIs.

- Risk-category summary.
- Workshop-region summary.
- Execution-status distribution.
- Top-priority vehicles.
- Full action report.
- Capacity, inventory, and data-dictionary sheets.

13.3 Model and governance report

The package includes model metrics, assumptions, validation controls, acceptance tests, and production-readiness recommendations.

14. Functional Requirements

ID	Requirement
FR-01	Load vehicle, workshop-capacity, and inventory data.
FR-02	Validate required columns, categories, ranges, and target values.
FR-03	Generate a failure probability for every valid vehicle.
FR-04	Apply safety overrides before probability thresholds.
FR-05	Generate action, timeframe, priority, required parts, and reason.
FR-06	Calculate expected breakdown cost, intervention cost, and expected savings.
FR-07	Allocate workshop slots in priority order.
FR-08	Consume parts inventory only when a slot is reserved.
FR-09	Produce parts and capacity escalations.
FR-10	Export detailed and summarized reports.
FR-11	Record enough information to audit why a decision was made.
FR-12	Capture outcomes for monitoring and future improvement.

15. Non-Functional Requirements

Area	Requirement
Explainability	Every action must have a decision reason and traceable inputs.
Reliability	Re-running with the same data and random seed must reproduce the result.
Performance	Batch processing must complete within the agreed operational window.
Security	Customer and vehicle identifiers must be access-controlled and encrypted where appropriate.
Privacy	Use only necessary customer attributes and enforce retention rules.

Auditability	Store model version, policy version, timestamp, and action status.
Availability	Escalation and safety decisions must remain available during partial-system failures.
Maintainability	Thresholds, costs, effectiveness assumptions, and capacity rules should be configurable.

16. Data-Quality Controls

- Vehicle ID is present and unique.
- Mileage, service gap, brake wear, and age are within agreed ranges.
- Battery state is either Low or Normal.
- Customer priority and workshop region use approved categories.
- Failure target is 0 or 1 for training data.
- Capacity and inventory values are non-negative integers.
- Duplicate events and stale records are flagged.
- Missing safety fields are sent to a manual-review queue rather than treated as low risk.

17. Testing and Acceptance

Test	Input / condition	Expected result
Safety override	Brake wear = 95%, probability = 0.45	Critical action within 24 hours
High risk	Probability = 0.72, no safety override	Priority workshop inspection within 48 hours
Medium risk	Probability = 0.50	Planned service within 7 days
Low risk	Probability = 0.25	Remote diagnostic and monitoring
No action	Probability = 0.10	Next regular service
Parts shortage	Required part stock reaches zero	Parts escalation, no false confirmed reservation
Capacity shortage	All near-term slots consumed	Capacity escalation
Invalid category	battery_voltage = Unknown	Validation error or exception queue
Reproducibility	Same data and seed	Same risk model and report order
Auditability	Any action row	Probability, rule, cost, constraints, and reason are traceable

18. Risks, Limitations, and Controls

Risk / limitation	Control
Synthetic training data	Replace with representative historical data and revalidate all metrics.
Incorrect cost assumptions	Obtain finance-approved cost and effectiveness values.
Probability calibration	Evaluate calibration curves and recalibrate before expected-

	value decisions.
Bias across regions or customer tiers	Compare error and action rates by segment; do not use customer tier to deny safety service.
Inventory data delay	Use reservation transactions and timestamps; reconcile with ERP.
Capacity changes after recommendation	Re-check availability at booking confirmation.
Automation error	Keep human approval for safety-sensitive or high-cost actions during rollout.
Model drift	Monitor feature, probability, action, and outcome drift.

19. Production Enhancement Roadmap

1. Replace the synthetic dataset with governed operational data.
2. Add probability calibration and cost-sensitive evaluation.
3. Store policies in a configurable decision table.
4. Integrate real-time workshop calendars and ERP stock reservations.
5. Add service-duration, technician-skill, bay-type, and customer-preference constraints.
6. Use an optimization solver for multi-period scheduling when needed.
7. Add API endpoints and event-driven action updates.
8. Add model registry, policy versioning, audit logs, and approval workflow.
9. Measure realized savings and failure avoidance, not only modeled expected savings.
10. Run controlled pilots before full automation.

20. Suggested Training Flow

Duration	Activity
15 min	Prescriptive analytics concept and 5W1H
15 min	Business problem, data, objectives, and stakeholders
15 min	Architecture and decision workflow
20 min	Python model and prescription logic walkthrough
15 min	Capacity, inventory, and escalation demonstration
10 min	Reports, KPIs, testing, risks, and production roadmap

21. Package Deliverables

- Case-study document in Markdown, Word, and PDF.
- Architecture and decision-flow diagrams.
- Vehicle input, workshop-capacity, and parts-inventory CSV files.
- End-to-end Python solution and requirements file.

- Detailed vehicle action report.
- Risk and workshop-region summaries.
- Excel dashboard.
- Executive report.
- Data dictionary.
- Requirements and acceptance-test document.
- README with recommended file order.

22. Final Business Interpretation

Descriptive analytics explains which vehicles failed and what patterns were observed. Predictive analytics estimates which vehicles are likely to fail. Prescriptive analytics closes the operational gap by deciding which action should be taken, when it should occur, what resources are needed, whether the action is feasible, and how exceptions must be escalated.